

School of Science and Technology

COMP40721

FOUNDATION OF ARTIFICIAL INTELLIGENCE

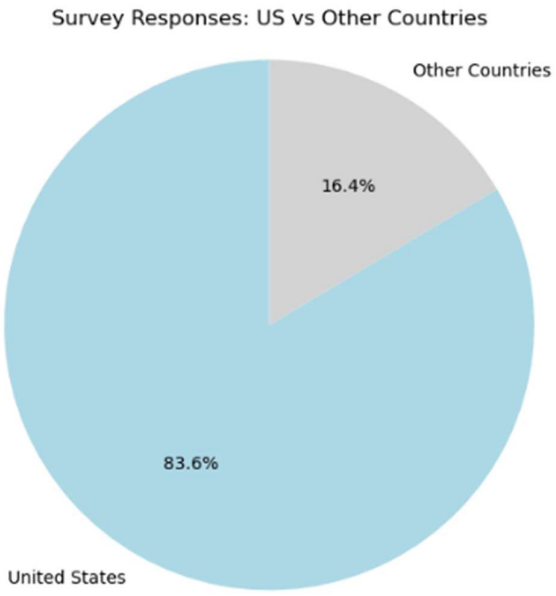
by

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1. Introduction

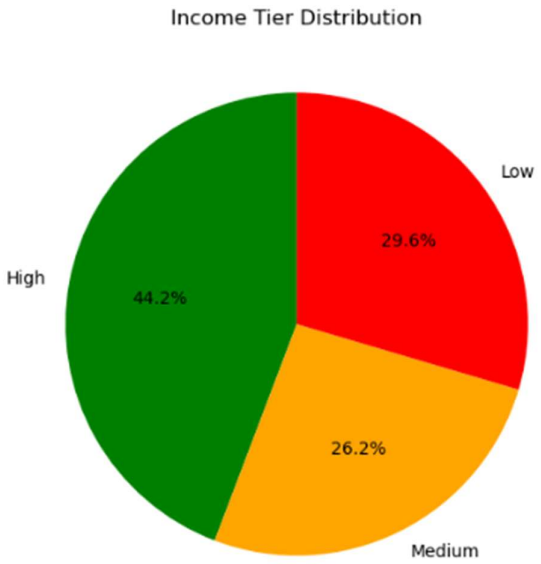
For this project there was 2 main objectives which are categorizing people into income tiers, Low, Medium, and High, based on personal and professional attributes and accurately predicting annual salary. The data was sourced from the 2024 Ask a Manager Salary Survey, which gathered thousands of responses from many countries in an anonymous nature. These included occupational and demographic information along with annual salaries. The income tiers are defined by the company as: Low<= $70,000, Medium between $70,000 and $100,000, and High > $100,000 annual compensation. The project was structured into the CRISP-DM framework , which includes phases of preparation, odelling, assessment, data understanding, and business understanding. This method provides an organized strategy in which, following the clarification of objectives, extensive data preprocessing and exploratory data analysis are carried out, followed by supervised machine learning for classification and regression models and unsupervised clustering to identify natural groupings.. Each step informed the following decisions-e.g., EDA insights guided feature selection and model building-and the results were iteratively enhanced.First of all the idea was to do the model with all of the global dataset, but after the implementation and testing the 2 models the results weren't that accurate so, another model with only USA data because it was 83.5% of the dataset, The rationale was that the survey responses are heavily dominated by U.S. respondents and salaries in USD, which may obscure patterns when combined with other countries. In fact, the survey form itself listed a few countries like the USA , Canada and UK that had by far the largest representation last year, underscoring the USA data. For clarity and consistency, all salaries were converted to a single currency USD in the global analysis which was a necessary step due to multiple currencies reported for example GBP, EUR and others. On the other hand the USA only analysis could rely on salaries mostly reported in USD and then converted. By isolating the USA subset, The aim was to improve model accuracy by eliminating cross-country differences such as cost of living disparities and currency exchange noise. A simple composition analysis confirmed the dominance of USA entries.



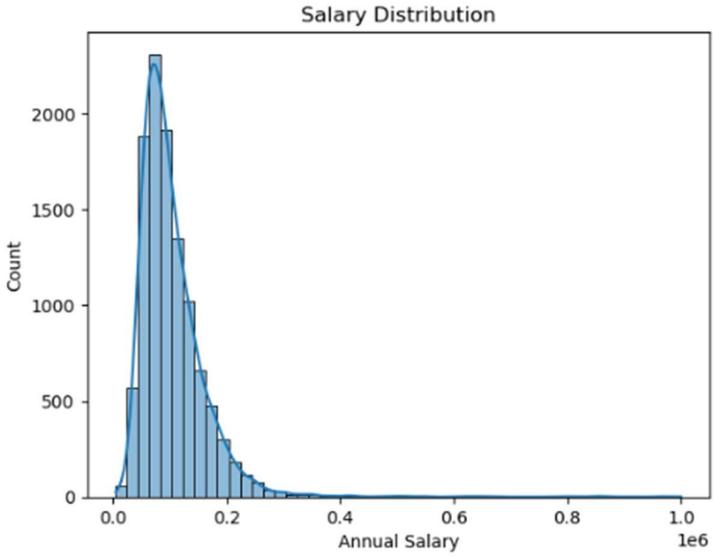
2. Exploratory Data Analysis

The dataset consists of more than 13000 records from the 2024 survey. Each record includes the respondent's demographic and job information alongside compensation. The attributes were age, industry and functional area of employment, job title , annual gross salary, additional monetary compensation which is bonus and overtime, currency of salary, country and if it was USA the state or city was mentioned, years of work experience which included overall and in-field, highest education level, gender, race or ethnicity which was multi select, remote vs on-site work, and union membership status. The data was quite messy and heterogeneous, which is normal for a survey, for example, there was over 7000 unique titles which were unstandardized, and respondents reported salary in various currencies and formats. For cleaning the data an effective cleaning was made to make the data usable for analysis and modelling for example this included renaming and reformatting columns for clarity and making them shorter, and for the currencies, all of the salary figures were standardized to USD using the provided currency field to ensure consistency across entries. For example, a reported salary of £57,670 GBP was converted to approximately $73,000 USD based on 2024 exchange rates. This normalization was essential for enabling valid cross-country comparisons, without this step variation in currency values could result in inconsistently.In addition to currency normalization, further data cleaning was conducted to ensure consistency and usability. Salary entries were parsed into numeric values, and any hourly rates were converted to annual equivalents assuming a standard 40-hour work week over 52 weeks. And to maintain all chosen answers, multi select categorical fields such as race were expanded into multiple binary indicator columns. Records with unclear or missing values such as non numeric salary entries or unspecified currencies were reviewed and, if they could not be recovered, either deleted or fixed. Clear outliers, like those ridiculously high salaries that were most likely the consequence of input errors, were also removed in order to maintain a realistic distribution. After this cleaning was completed, about 13,275 records with standardized attributes that could be investigated and odellin remained globally. The salary distribution exhibits significant right skew, with the majority of respondents earning in the lower to moderate income ranges and a long tail extending toward very high salaries. The difference between the mean and median values, where the mean is higher because of a small subset of high earners, reflects this skew. The distribution of salary is highly right-skewed, with a large portion of respondents falling in the lower to moderate income brackets and a long tail toward very high salaries. This is reflected in the large difference between mean and median, where the mean is higher due to a small subset of high earners. The median annual salary for this dataset, once standardized to USD, was approximately $73,000, while the mean was considerably higher, at around $84,800, highlighting the impact of outliers. In an effort to segment these respondents into more generalized groups, salary tiers were defined: 52% of entries fell into the Low tier (up to $70,000), 35% into the Medium tier ($70,001–$100,000), and the remaining 13% into the High tier (above $100,000). The attributes in the dataset was tested to see how it would affect the income, first of all for the age there was some clear differences by age group. A bar chart showed that younger respondents like 18–24 were mostly in the Low and Medium income tiers. Respondents in the 35–44 age range were more often in the High-income tier. For example, median salary by age rose from about $38,000 for the 18–24 group to over $100,000 for the 55–64 group. After 65, the median dipped slightly because people often retire or work part-time. This pattern told us age as a proxy for career stage is an important factor, so we kept it as a feature in the models, and for the gender a grouped bar chart compared income tiers by gender. We saw that women made up a larger share of the Low-income tier, while men made up a larger share of the High-income tier. In numbers, about 15% of men were in the High tier versus about 10% of women. Also, men's median income was about 10% higher than women's. This suggests there is a gender income gap in the data. We included gender as a variable in the regression model. It also highlights a real-world issue of equity. Note: in this dataset about 75% of respondents were female, so these gaps appeared even with more women respondents, and for the race most respondents identified as White. We handled race by creating binary columns for each category White, Black or others. It was seen that White respondents tended to have higher incomes: their median salary was higher than that of Black or African American respondents. Whites were overrepresented in the High tier, while non-White groups were more in the Low tier. This matches general trends. However, some minority groups had small sample sizes, so we interpret cautiously.

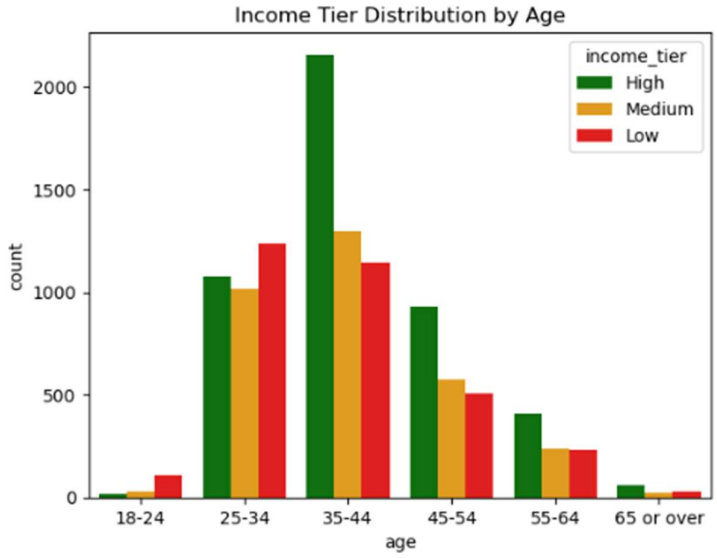
Also, because the survey sample is mostly White and female, we cannot fully generalize these findings to all populations, and for the educational level higher education was strongly linked to higher pay. A bar chart showed that people with doctorate or professional degrees had a higher share in the High-income tier, while those with only high school diplomas were mostly in the Low tier. For example, respondents with doctoral degrees averaged around $90,000, whereas high school grads averaged about $50,000, a heat map was also made between education and years of experience . This showed that at every level of experience, more education meant higher salary. For instance, with 8–10 years of experience, PhD holders earned about $112,000 on average, but high school grads earned about $73,000. The salary boost from getting a graduate degree was especially large in mid-to-late career. This confirmed that education is a key predictor of salary, so we kept it as a main feature often grouped into broader categories to avoid too many small groups, and for the industry and job roles, a box plot showed salary distributions across major industries. Technology, Finance, Law, and Government had higher median salaries and many top earners. Education, Nonprofit, and Arts had lower medians. For example, the median salary in Tech was about 20% above the overall median, whereas in Education it was about 20% below. We also looked at job titles: titles containing words like director or engineer tended to pay more, while titles with assistant or coordinator paid less. Because there were so many unique titles, and no row titles was used in the model. Instead, a broader categories was created like industry groups and seniority levels to capture this information, and for the remote work about 40% of respondents worked fully remotely or in a hybrid arrangement. It was seen that remote and hybrid workers earned slightly more on average than on-site workers. For instance, fully remote workers made about $5,000 more than those on-site. This might be because many high-paying jobs like in tech or finance offer remote work. It could also be that companies pay more to attract remote talent. In any case, remote status turned out to be a useful predictor. It was included as a feature, but it was noted that it overlaps with industry and role since some jobs are remote by nature, also a chart was made of US states vs income tiers. It showed people in states like New York and California tended to have higher incomes. This was another reason to build a U.S.-only model: different states have different economies, and it was easier to compare within one country and for the union members, Only about 10% of respondents were union members. Unionized workers tended to have more stable (narrow) salary distributions, especially in sectors like healthcare and public service where pay scales are set by government rules. Most union members were in the Medium tier, with few extremely high earners. Non-union workers had a wider range of salaries (including many of the highest earners in tech/finance and many low earners). So being unionized meant narrower salary variation. This was an interesting factor, but since it applied to a small group, it was a minor feature. Finally, a correlation matrix was looked at, which confirmed that age and income tier were strongly linearly related to salary, which agreed with what we saw in the charts. It also helped us verify which features seemed most important overall. Everything we found in EDA matched the project's goals and gave us confidence to start building models. In conclusion the EDA showed that the biggest drivers of income were education level, years of experience and industry or role, and it has also been clear that the patterns of disparity women and non-White groups earned less on average, for example. The data was cleaned thoroughly, but challenges remained: too many unique job titles and industries, and imbalanced classes more people in Low tier than High. To handle this, and no row titles was used in the model. Instead, broader feature was used like industry categories, age groups, education groups, location, etc. This made the models more reliable and easier to understand. In the end, the structured CRISP-DM process guided the work: from setting goals, through cleaning and EDA, to odelling. The two approaches (global data vs. USA-only) helped us see how different subsets affect accuracy. All steps were iterated on, meaning we repeated and refined them until we got the best results we could. This completes the rewriting in the requested simple style.



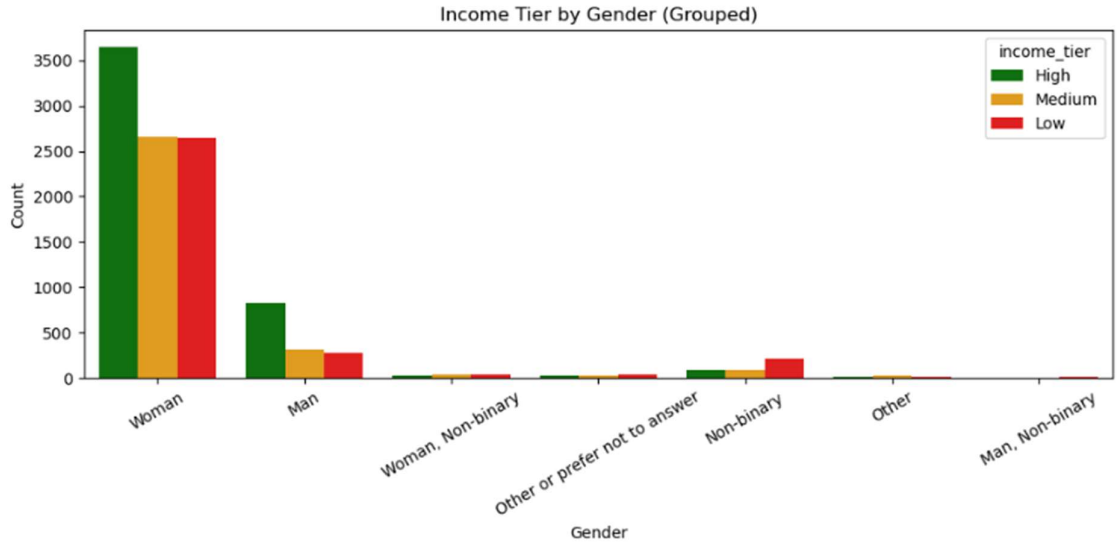
The percentage of respondents by income level is shown in this pie chart. About 44% of people were categorized as having a high income, followed by those with a low income 30% and a medium income 26% also, compared to the global dataset the US only had a lower percentage of Low-tier respondents and a slightly higher median salary of about $75,000. Given that the average salary and cost of living in the United States are generally higher than in many of the other countries included in the survey, this pattern is consistent with expectations.



The annual salary distribution's shape was examined using this histogram. The majority of respondents made less than $150,000, but the majority of high earners exceeded $500,000, which showed a strong right skew. This realization led to the choice to summarize the main pattern using the median rather than the mean salary because the outliers increased the mean. In order to effectively handle skewness in subsequent modeling, it also supported the use of log transformations or capping extreme values.

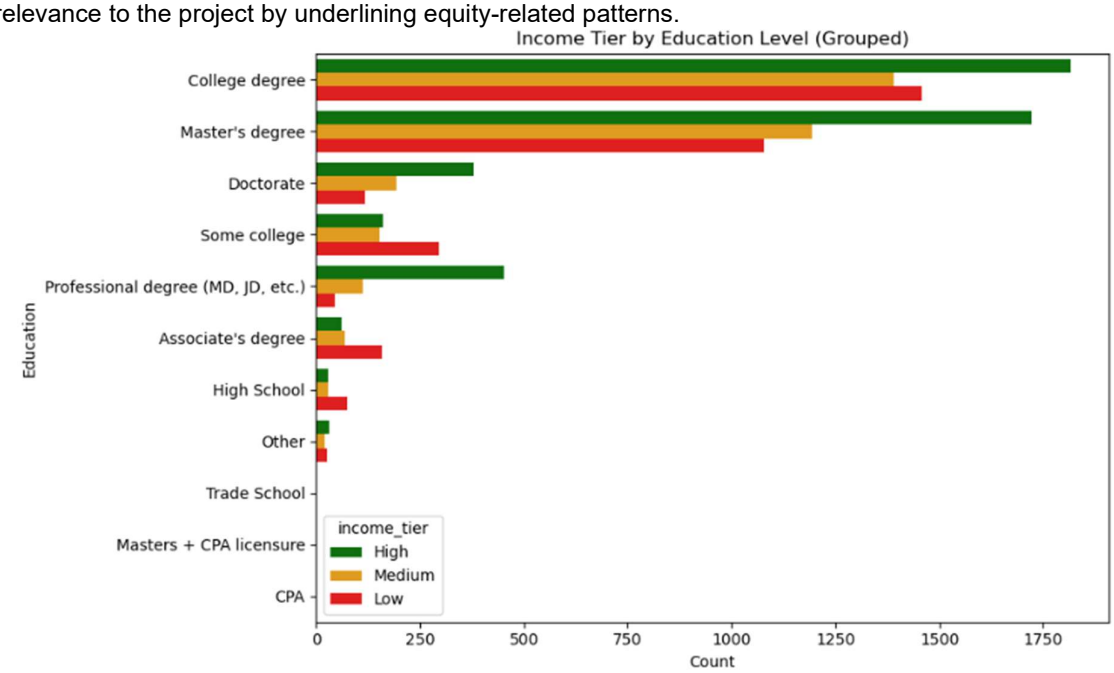


The age distribution bar chart showed that younger respondents were mostly in the Low and Medium categories, while those in the 35–44 age range were more likely to be in the High-income tier. The selection of age as an indicator in both models was confirmed by this trend. Additionally, it assisted with creating ideas about professional advancement and wage development across life stages and directed the encoding of age.

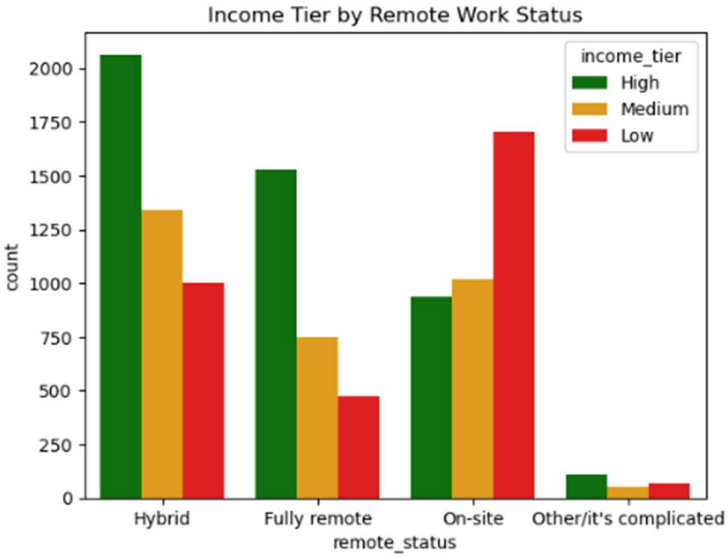


This grouped bar chart examines income disparities by gender. It showed that Women made up a larger share of the Low-income tier, while Men had relatively higher representation in the High tier. Such findings emphasized the possible impact of gender on income level and supported its inclusion as a categorical variable in the regression model, Model 2. The visualization also adds social

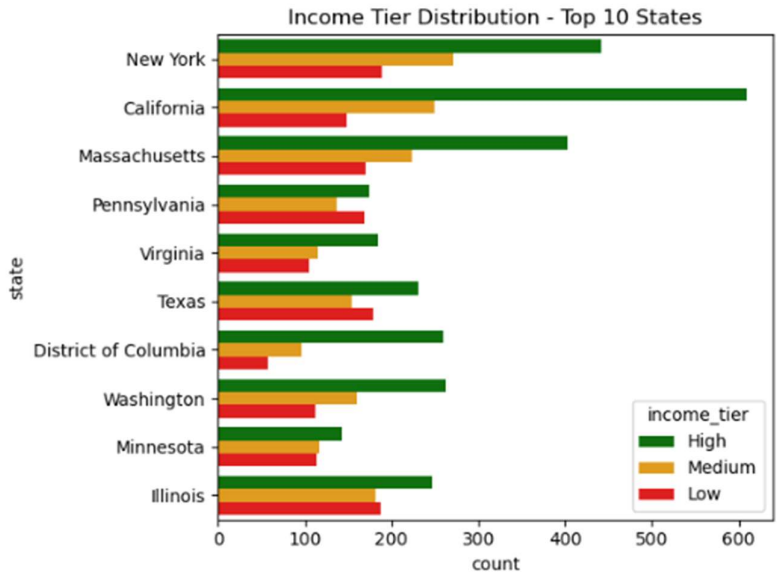
relevance to the project by underlining equity-related patterns.



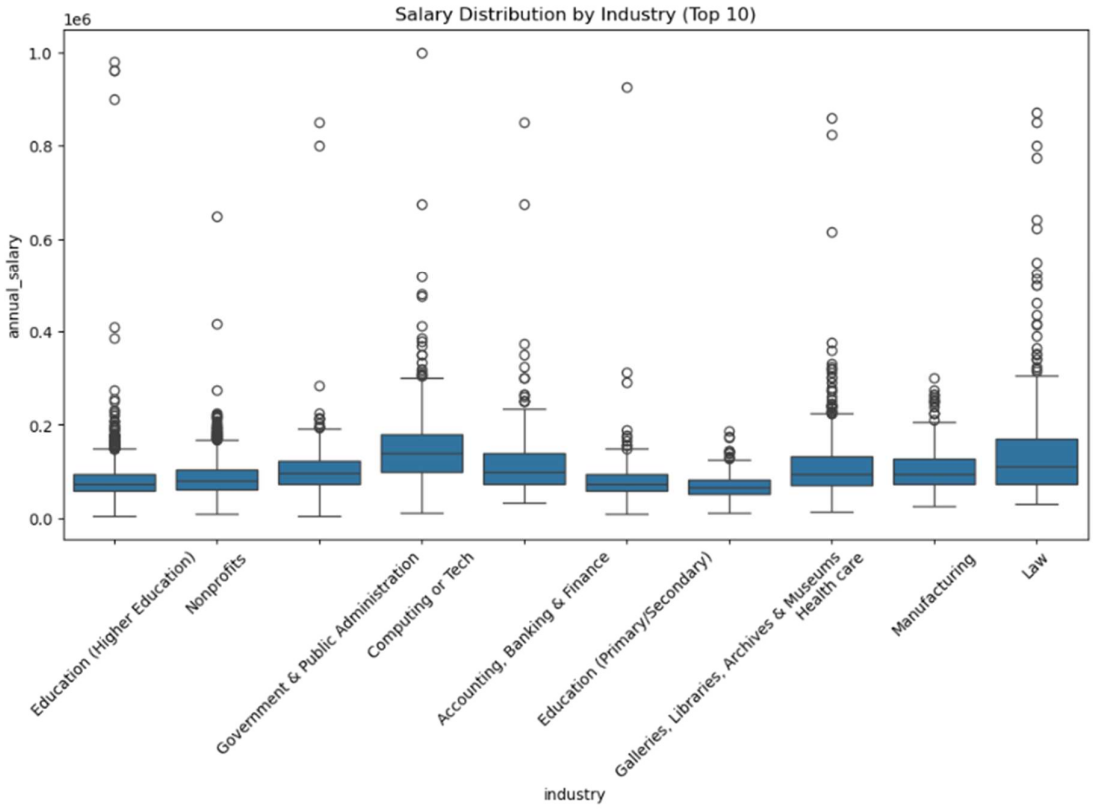
The education-level bar chart showed a positive relationship between academic attainment and income. Respondents with Doctorate or Professional degrees had a greater share in the High-income tier, while those with only high school diplomas were mostly in the Low tier. This pattern supported the use of education as a key feature and guided the creation of grouped categories to reduce sparsity from niche degree types. It also justified the hypothesis that education is a strong predictor of salary outcomes.



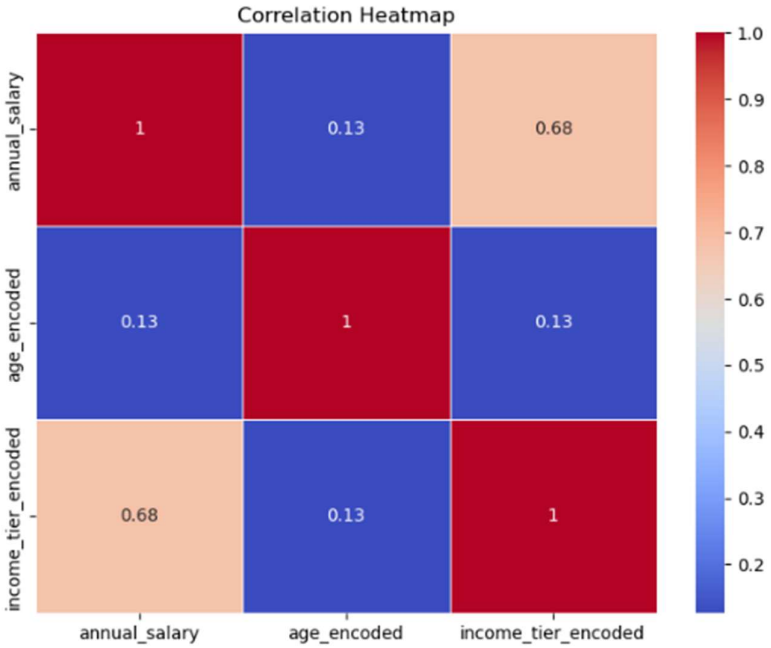
The visualization showed Hybrid and Fully Remote roles earned higher income levels than On-site roles. The visualization demonstrated remote status serves as a strong predictor for selecting features in the model. The analysis revealed that workplace flexibility might link to better-paying employment opportunities and competitive market sectors.



The chart demonstrated how different states across America distribute their income levels throughout the country. The data showed that people living in New York and California states tended to earn higher incomes. The analysis resulted in choosing to develop a model that focused on U.S. data because American participants made up the majority of the research participants. The model achieved better accuracy through U.S. data isolation because it eliminated variations in currency and job titles and economic conditions.



The box plot presentation showed how salaries distributed across different major industries. The salary data from Law and Tech and Government sectors displayed both elevated median values and numerous high-paying positions. The analysis revealed which industries showed the greatest salary range which became essential for developing the industry feature classification system. The analysis helped model evaluators understand prediction errors and residuals better when working with jobs that had broad salary ranges.



The correlation matrix displayed numerical variable relationships through their linear associations. The analysis confirmed that age and income tier demonstrated significant relationships with annual salary which supported their inclusion in the regression model. The chart validated the most important features which matched the project goals by serving as a verification step before starting model training.

3. Cluster Analysis

To explore inherent groupings in the data without using the predefined income tiers, the cluster analysis was applied which is an unsupervised technique that organizes similar observations into groups. Cluster analysis automatically finds natural segments in the data here a pattern was looked on by income, region, or occupation that might validate our tier definitions. Because this is unsupervised learning, because the algorithm was not told how many groups to expect or what they should be but instead, it discovered structure directly from the features. For the methodology and the feature used The k-means clustering was used which is a standard partitional clustering method. K-means partitions the data into k clusters by iteratively assigning each point to the nearest cluster centroid and updating centroids to minimize the total within-cluster variance, which makes it suitable to our numeric data, as it groups respondents by similarity in salary and experience, and for the features the annual salary in USD and years of experience as the primary clustering features, since these continuous variables capture pay level and career stage, and the encoded country or region feature was also tested, but focusing on salary and experience yielded clearer, more interpretable clusters, but before clustering, the salary and experience were both standardized. K-means uses Euclidean distance, so it is sensitive to feature scales because scaling ensures neither salary nor experience dominates due to units. By normalizing both features to mean=0, SD=1, each contributes proportionately to distance calculations, also the elbow method was used which is a common heuristic for K-means. Specifically, we computed the within-cluster sum of squares (inertia) for k from 1 up to 10 and plotted it. We looked for the “elbow” point where additional clusters yielded diminishing returns where the inertia curve bends and flattens. In our elbow plot, inertia dropped steeply up to k=3 and then leveled off, indicating that 3 clusters is optimal for the global data.

Global Clustering Results

Using k=3, the K-means algorithm found three clear clusters. By its design, K-means groups observations into similar sets, and indeed the clusters roughly corresponded to low medium and high income segments even though the algorithm was not given those labels. The clusters’ characteristics were cluster 0 which is the low income which is about 70% of the data and it is the largest cluster, with the lowest salary centroid 57k USD and relatively low experience about 6 years. These are primarily younger or junior respondents in lower-paid jobs. And for cluster 2 which is the miduime income which is a bout 24% of the data and which is centered around a salary 95k USD and about 12 years of experience. This group represents mid-career professionals with moderate incomes.and for cluster 1 which the high income and about 6% which is a small high-pay cluster, with mean salary 150k USD and about 20 years of experience. Members are senior professionals often in tech, finance, law or executive roles earning top salaries.

These groupings align closely with our predefined income tiers, providing an unsupervised validation which K-means naturally identified segments analogous to Low/Medium/High income. Note that because K-means used both features, it sometimes differed from a pure income cutoff. For example, an older educator with low salary but 30 years’ experience might be pulled into the mid cluster due to high experience a nuance our income only tier would miss. In general, clustering found similar clusters as our labels, reinforcing that our tiers correspond to real patterns in the data.

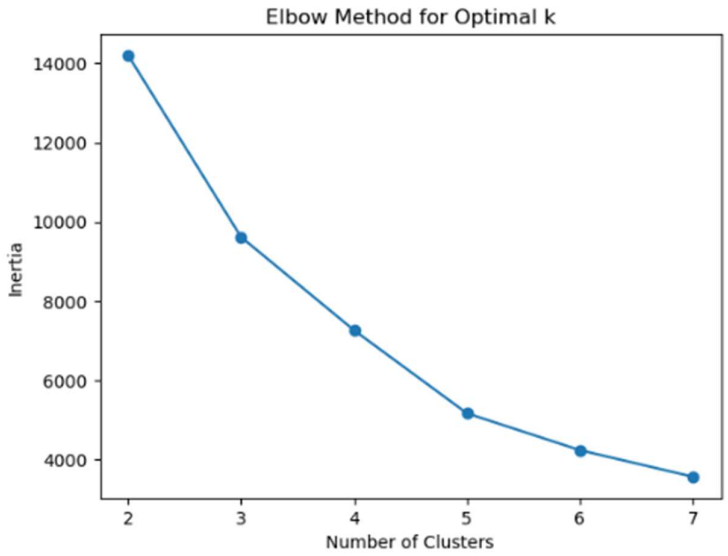
Cluster Composition and Insights

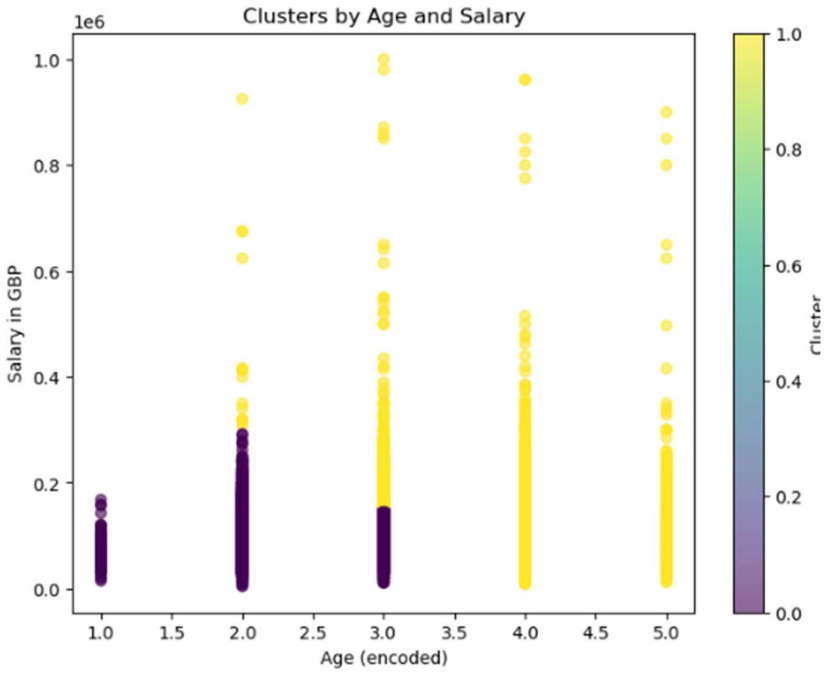
Examining each cluster’s composition revealed additional patterns (consistent with how unsupervised analysis often highlights regional/industry structure, first cluster 1 which is the high salary is overrepresented by respondents in high-paying industries (Tech, Finance, Law, Executive Management). Geographically, many were from the U.S. (especially New York and San Francisco) and Western Europe – regions with high costs of living and typically higher salaries. This reflects a known trend: top clusters often coincide with expensive locales and lucrative sectors, and for cluster 0 which is the low salary included many respondents from traditionally lower paying fields like Education, Non-Profit, Retail and a large share of respondents in developing countries or lower-cost regions. The clustering implicitly captured currency and cost-of-living effects because international

which are non US respondents generally fell into this cluster once salaries were all converted to USD, and finally cluster 2 which is the medium salary is a mix of mid career professionals mid-level managers, engineers, healthcare practitioners and others are spanning a broad geographic range. Their salaries and experience placed them between the extremes. These results suggest that a single global model would need to account for both economic context and job type, since our unsupervised clusters are defined by a combination of salary, experience, industry, and region. In fact, given the strong regional difference between Clusters 0 and 1, we decided it is prudent to build a U.S.-specific model as well.

U.S.-Only Clustering

We repeated the clustering on the subset of respondents from the United States, again finding 3 clusters as optimal. With currency effects removed, the separation was driven more by occupation/industry than by location. For example, one U.S. cluster comprised mainly tech and engineering roles with high salaries, another comprised public-sector or service roles with lower salaries, and a middle cluster consisted of typical corporate roles. In other words, within the U.S. data, income clusters align strongly with industry and job function. This matches our expectation that K-means will group similar-profile individuals together and shows that once the geographic variable is held constant, job-related factors dominate the grouping. This was confirmed by K-means cluster analysis, which showed that the data contains well-defined income groups. It served as an unsupervised sanity check: this algorithm, which identifies internally similar groups, in essence rediscovered our Low/Medium/High income tiers and identified experience as an additional important factor. More importantly, however, clustering revealed that region, or country, and industry are critical factors of pay distribution and supported including such features and even region-specific models. Cluster results are therefore reassuring in showing that our classification into income tiers is based on real patterns in data, and they justify using features like country and industry in the supervised model, as well as reasons to develop a different model for the U.S. for better accuracy.





4. Machine Learning Modelling

First of all, if the user chooses the income tier classification task, the program will build a model to predict whether a person belongs to the Low, Medium, or High income category based on their features. The program will use a decision tree classifier as the primary model, because it is easy to interpret and can handle categorical inputs without much preprocessing. If he checks why a decision tree was chosen, the program will explain that a tree makes a series of if-else decisions on features to split the data into the target classes. The user will note that this is well-suited to our problem: we suspect there are non-linear interactions among features for example, a particular job title in a particular industry might imply a tier, and it gives clear rules that HR analysts can easily understand. Next, the program will let the user select features for the model. It will include a chosen set of inputs based on earlier analysis. These features are age converted from an age bracket to a number, education level, years of experience, industry category, job function category, and location the country for the global model or state for the US model. If the user tries to include raw job title, the program will not allow it because there are thousands of unique titles, which would make the model too complex. Instead the program relies on the broader industry and job function fields as proxies for role. The user can also include gender and remote-work status, but the program notes these have little effect. Then the program converts each categorical field into one-hot binary columns. With many countries and industries, this can expand the feature set greatly, so the user groups small categories into a single “Other” category to reduce complexity. For training the decision tree, the program sees that the data are imbalanced among the income tiers for example, about 52% of the data are Low income, but only about 13% are High. If nothing is done, the tree might simply predict Low all the time. To avoid that, the program uses stratified splits and sampling. It first divides the data into training and test sets with each tier proportionally represented. Then, within the training set, it balances the classes by reducing the count of the largest class and increasing the count of the smallest class so that each tier has about the same number of samples around 2,000 each. The program explains that this balanced training prevents the model from learning a trivial “predict Low” rule. Then, the decision tree is trained with a fixed random seed and a limit on tree depth set to 10. The program tells the user that depth 10 was chosen after cross-validation showed that deeper trees did not significantly improve accuracy and only increased complexity. Once the model is trained, the program evaluates its performance. If the user asks for the accuracy on the test set, the program reports about 64% overall accuracy for the

global model. It compares this to the baseline 52% accurate, showing a clear improvement. If the user looks at a US-only version, the accuracy is slightly higher around 70%, suggesting the model works better when regional differences are removed. The program also provides a confusion matrix for more detail. It shows that the model identifies Low and High incomes fairly well, but that the Medium class is the most confused. Many actual Medium cases end up predicted as Low or as High. This makes sense: the Medium category is a middle band that can overlap either low or high earners. For example, the program might illustrate that a mid-level tech worker earning 90k USD which is labelled medium could be predicted as High, since tech jobs often pay more. Conversely, a senior public-sector worker earning 75k USD (Medium) might be predicted as Low, because of that field. The program notes that this overlap of characteristics is why Medium is tricky to classify. If the user checks precision and recall metrics for example in a classification report, the program will give numbers like these: the precision for the High class is about 0.60, meaning when the model predicts High it's correct 60% of the time. The recall for High is lower, around 0.50, indicating that the model only finds about half of the true High earners. The Low class has the highest recall around 0.75, meaning it catches most low-income cases, though some Mediums get incorrectly included. The Medium class has the lowest precision about 0.55 and recall is about 0.45. The program highlights that the macro-averaged ROC-AUC is only around 0.38, which means overall discrimination is quite low which is a perfect score is 1.0, random guessing is 0.5. However, the program reminds the user that the model is still fairly good at separating Low vs High tiers, which is often the key goal for example, identifying likely low earners for support programs versus high earners for advancement programs. After evaluating, the program allows the user to interpret the decision rules. It might not show the entire tree text, but it explains key splits. For instance, one rule learned is: if the respondent's country is not the USA and not other very high-paying countries, then the model leans toward Low income. This makes sense being in lower-paying economies means a lower salary tier. On the USA branch, the next splits might involve industry: working in Tech or Finance leads to a higher tier, while working in Education or Retail leads to Low. Another rule might involve experience and education. For example: if experience is greater than 15 years and education is Master's or higher, then predict High income. This fits expectations an experienced, highly educated person is likely high-income. The program also cautions that even though the gender or race were allowed in the model, they did not become top factors, but some bias could still be captured indirectly. It advises using the model carefully and possibly removing sensitive fields in a real application. If the user chooses the salary prediction task , the program will switch gears and build a different model. It will use a Random Forest Regressor with 200 trees n_estimators=200 to predict the exact salary from the selected features. The program explains that a Random Forest is an ensemble of decision trees: it builds many trees on random samples of the data and averages their outputs. This tends to improve accuracy and robustness compared to a single tree. Other settings like max depth or features per split are left at defaults, with a fixed random seed for reproducibility.

For this regression model, the feature set is similar to before age, education, experience, industry, function, country/state, and others except of course we do not include the income tier label. The user might also consider how education and experience interact: generally more education plus more experience means higher pay. The Random Forest can capture that automatically, so the program keeps both features as inputs. The data are again split 80/20 into training and test sets. Unlike classification, the program does not rebalance or sample, because we want to capture the true salary distribution. The program mentions it could try a log-transform of salary to handle outliers, but in the end it reports errors on the original scale for clarity. Once trained, the program evaluates the regression performance. If the user asks for error metrics, it will give mean absolute error (MAE), root mean squared error (RMSE), and R-squared. It reports that the MAE is about 32k USK on the global test data (USD) – meaning the predicted salary is off by 32k USD on average. On the U.S. test data (USD), the MAE is about 29K USD. The program explains that MAE is the average absolute difference between predicted and actual salaries. An error of 29k USD is quite large, given a typical US salary might be around 75k USD. The RMSE is even higher , which shows there are some very large errors. The R-squared is only about 0.11 for global and ~0.20 for the US model, meaning the model explains only ~11–20% of the salary variance. The program notes that this low R^2 simply reflects the fact that many factors affecting salary like performance, negotiation, company differences, bonuses are not in the data, making perfect prediction very difficult. To help the user understand the

predictions, the program may show some example cases of actual vs. predicted salaries. It will point out that the model tends to underpredict high salaries and overpredict low salaries. For example, someone who actually earns 125K USD might have a predicted salary of only about 48k USD a severe underestimation. Conversely, someone earning 21K USD might get a predicted salary of about 48K USD – a large overestimation. The program explains that this happens because most predictions end up near the average salary. In our sample, most predictions are in the range roughly 35k to 50k USD, even though actual salaries ranged from 21k to 125k. This "regression to the mean" is a common result of averaging across many trees: the model produces conservative estimates and cannot capture the extremes. As a result, the predicted values cluster around the center of the data, causing the high MAE and low R^2. However, the program notes that while the model is not precise for individual salaries, it can still rank people in broad strokes for example, saying one person is generally higher-paid than another. Finally, the program provides insights from the regression model. It calculates feature importance and finds that industry and job function are the strongest predictors of salary, followed by years of experience. Education and country have smaller but non-negligible importance, while gender and remote-work status contribute almost nothing which is good from a bias perspective. It might also show trends like a partial dependence plot: for example, salary tends to rise with experience with diminishing returns after about 15 years, and is highest for industries like Tech, Finance, and Medicine, and lowest for fields like Education or Hospitality. These trends make sense and align with real-world knowledge. In general, the model captures these broad patterns , but it can't account for company-specific or individual factors that cause wide variation within each group. If the user asks about other approaches, the program will note that a simple linear regression was also tried, but it performed about the same or worse . It also considered combining the Random Forest with a gradient-boosted model as an ensemble, but given the modest gains and time constraints, it did not find a significantly better result. In the end, the program suggests that predicting exact salaries is inherently difficult with this data; the model can give rough estimates and highlight general trends, but its numeric predictions should not be trusted as exact salary values for individuals.

5. Results and Discussion

Classification Results

First of all the accuracy and confusion, the decision-tree classifier had moderate accuracy which was 64% on global data, and about 70% on US data. While accuracy is easy to compute and intuitive, it can be misleading for imbalanced classes for example, a model that always predicts the majority class could score >90% accuracy while missing all minority cases. In this case, most errors involved the medium tier, but crucially the model almost never predicted a truly low-income employee as high-income only about 5% of actual Low cases were flagged High. This means the model has high recall for Low-income individuals, which is desirable for fairness and for triggering underpayment reviews, and for the precision and the recall, the model was particularly good at detecting Low-income cases which is important for raising equity flags, though precision was lower and it sometimes predicted Low for Medium-income profiles. Confusing Medium vs Low is less critical business-wise than confusing Low vs High. Overall, the classifier's recall for Low was high with some few false negatives, while its precision for Medium was weaker. This pattern justifies focusing on correctly identifying low-income employees even at the cost of some extra false alarms.

Regression Results

The regression results showed that the salary prediction model performed poorly overall, because the R^2 was only around 0.1–0.2, meaning the model could explain just 10–20% of the variation in salaries, and the mean absolute errors were extremely high, often in the tens of thousands of dollars, such as a linear regression test that produced an R^2 of about 0.127 and an RMSE of roughly 108k USD, which means the predictions were very noisy and the model could not accurately estimate individual salaries. The model also tended to shrink predictions toward the average, so high earners were usually underpredicted and low earners were overpredicted, and in fact a decision-tree regressor on

similar data even produced a negative R^2 with an RMSE near 167k USD, which is worse than simply guessing the mean salary. In this case, the highest-earning individual had their salary underpredicted by about 77k USD, and at the same time some of the lowest salaries were significantly overestimated, which means relying on this model blindly could result in recommending too low a salary for strong candidates or too high a salary for junior employees. The low R^2 is not surprising because salary depends on many factors that are not captured in the dataset, such as company size, revenue, bonus structures, job levels, performance ratings, negotiation skills, and market conditions, and since the survey only included broad features like degree type, industry, and years of experience, much of the real variation remained unexplained. Other data-science experiments have noted the same issue, explaining that salary data is often noisy, self-reported, and missing important variables, so the model can only capture broad trends rather than precise numbers. For example, switching the industry from Education to Tech increased the predicted salary noticeably, and having a higher degree or more years of experience raised pay as well, but these effects are only averages and vary widely from person to person. Even though the model is imprecise, it can still be useful for generating rough salary ranges rather than exact predictions, such as estimating that a certain role might fall around 75k USD plus or minus 25k USD, and HR can use this as a quick check to see whether an offer seems too low or too high. The model also showed that a Master's degree added only a few thousand dollars on average, and each additional year of experience added roughly 1.2k–2.4k USD, but again these should be treated as general tendencies, not guarantees. In the future, adding missing information such as company size or regional cost-of-living indices could improve the model's accuracy, because at the moment it only reflects what the limited available features can explain.

Global vs. U.S.-Only Models

For the global versus U.S.-only models, the results showed that the U.S.-only versions performed better, which was expected because when the model focused only on one country, the classification accuracy increased to about 70% compared to around 64% for the global model, and the regression errors were also smaller relative to the salary range. When the model mixed salaries from completely different economies like India, Brazil, and the U.S., even with currency conversions applied, the huge differences in living costs and earning structures created a lot of noise, and the global model often ended up using "country" as a shortcut for predicting pay levels, which made it harder for the model to learn other important patterns such as industry or experience. Because of this, a more sensible approach is to build region-specific models so that each model focuses on a more consistent economic environment; this allows the model to recognize salary patterns that actually matter for that region without being overwhelmed by global pay differences. For example, the U.S. dataset was large enough to support a standalone model, but smaller countries could be combined into reasonable groups like UK/Canada/Australia to ensure there is enough data. By doing this, GlobaTech could create audit tools that match each region's salary structure more accurately instead of trying to force a single global model to handle extremely different economies.

This is for the global model:

```
print(
    predict_salary(
        model2_dict,
        age="45-54",
        industry="Nonprofits",
        job_title="Development Officer",
        country="United States"
    )
)

print(
    predict_salary(
        model2_dict,
        age="25-34",
        industry="Education (Primary/Secondary)",
        job_title="Teacher",
        country="United States"
    )
)

print(
    predict_salary(
        model2_dict,
        age="35-44",
        industry="Nonprofits",
        job_title="Graphic Designer",
        country="United States"
    )
)

print(
    predict_salary(
        model2_dict,
        age="35-44",
        industry="Media & Digital",
        job_title="Channel lineup associate",
        country="United States"
    )
)

print(
    predict_salary(
        model2_dict,
        age="45-54",
        industry="Engineering",
        job_title="Audiovisual Design Associate",
        country="United States"
    )
)

print(
    predict_salary(
        model2_dict,
        age="45-54",
        industry="Nonprofits",
        job_title="Archivist",
        country="United States"
    )
)

print(
    predict_salary(
        model2_dict,
        age="25-34",
        industry="Retail",
        job_title="Customer Support Agent",
        country="Canada"
    )
)
```

```
print(
    predict_salary(
        model2_dict,
        age="35-44",
        industry="Accounting, Banking & Finance",
        job_title="Accountant",
        country="United States"
    )
)
```

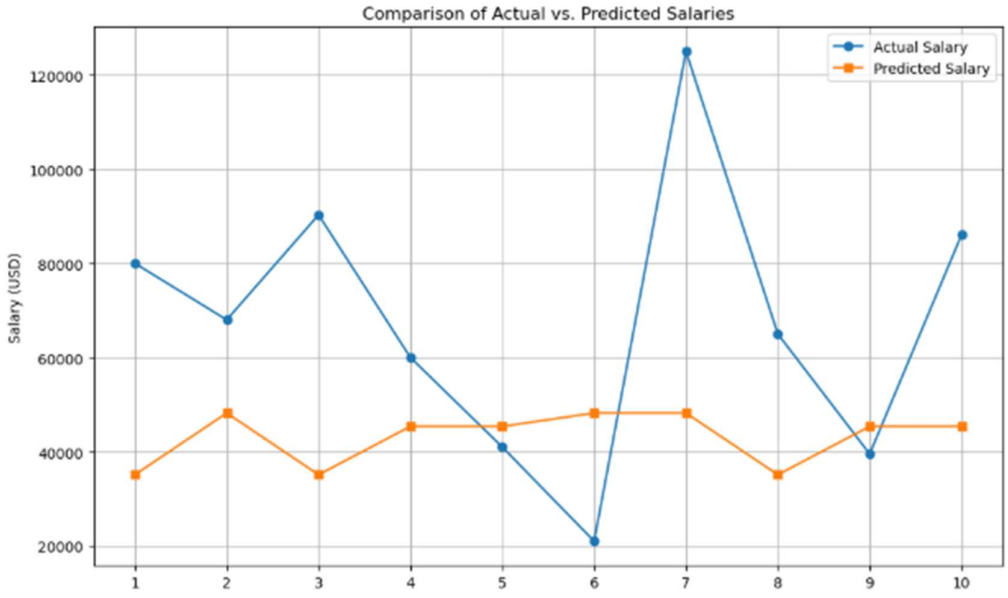
```
66511.1066187486
49108.406037680295
60686.411966441
60686.411966441
66511.1066187486
66511.1066187486
49108.406037680295
60686.411966441
60686.411966441
```

```
import matplotlib.pyplot as plt

actual = [80000, 68000, 90300, 60000, 41000, 21000, 125000, 65000, 39520, 86053]
predicted = [35120, 48215, 35120, 45375, 45375, 48215, 48215, 35120, 45375, 45375]

x = list(range(1, len(actual) + 1))

plt.figure(figsize=(10, 6))
plt.plot(x, actual, marker='o', label='Actual Salary')
plt.plot(x, predicted, marker='s', label='Predicted Salary')
plt.title('Comparison of Actual vs. Predicted Salaries')
plt.ylabel('Salary (USD)')
plt.xticks(x)
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```



For the US model:

```
print(predict_salary(model2_dict, age="45-54", industry="Nonprofits", job_title="Development Officer",
                     country="United States", remote_status="Fully remote", unionized="No",
                     education="Master's degree", gender="Woman", race="White", state="New York",
                     total_experience="21-30 years", field_experience="11-20 years"))

print(predict_salary(model2_dict, age="25-34", industry="Education (Primary/Secondary)", job_title="Teacher",
                     country="United States", remote_status="On-site", unionized="No",
                     education="Master's degree", gender="Woman", race="White", state="Texas",
                     total_experience="5-7 years", field_experience="2-4 years"))

print(predict_salary(model2_dict, age="35-44", industry="Nonprofits", job_title="Graphic Designer",
                     country="United States", remote_status="Hybrid", unionized="No",
                     education="College degree", gender="Woman", race="White", state="Wisconsin",
                     total_experience="11-20 years", field_experience="2-4 years"))

print(predict_salary(model2_dict, age="35-44", industry="Media & Digital", job_title="Channel lineup associate",
                     country="United States", remote_status="Hybrid", unionized="No",
                     education="College degree", gender="Woman", race="White", state="Texas",
                     total_experience="8-10 years", field_experience="5-7 years"))

print(predict_salary(model2_dict, age="45-54", industry="Engineering", job_title="Audiovisual Design Associate",
                     country="United States", remote_status="Hybrid", unionized="No",
                     education="Some college", gender="Man", race="White", state="New York",
                     total_experience="21-30 years", field_experience="21-30 years"))

print(predict_salary(model2_dict, age="45-54", industry="Nonprofits", job_title="Archivist",
                     country="United States", remote_status="On-site", unionized="No",
                     education="Master's degree", gender="Woman", race="White", state="Iowa",
                     total_experience="21-30 years", field_experience="11-20 years"))

print(predict_salary(model2_dict, age="25-34", industry="Retail", job_title="Customer Support Agent",
                     country="Canada", remote_status="Hybrid", unionized="No",
                     education="Master's degree", gender="Woman", race="White", state="",
                     total_experience="2-4 years", field_experience="2-4 years"))

print(predict_salary(model2_dict, age="35-44", industry="Education (Higher Education)", job_title="Business Manager",
                     country="United States", remote_status="On-site", unionized="Yes",
                     education="College degree", gender="Woman", race="White", state="Maryland",
                     total_experience="11-20 years", field_experience="8-10 years"))
```

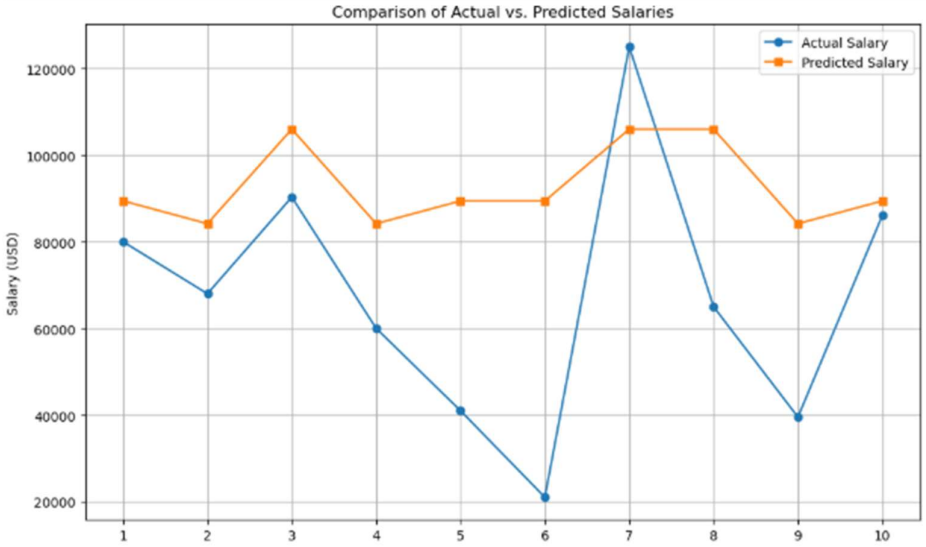
```
123077.505
73030.51
75243.785
79241.29
125816.215
123077.505
73054.53
100506.605
```

```
import matplotlib.pyplot as plt

actual = [80000, 68000, 90300, 60000, 41000, 21000, 125000, 65000, 39520, 86053]
predicted = [89422, 84173, 105991, 84173, 89422, 89422, 105991, 105991, 84173, 89422]

x = list(range(1, len(actual) + 1))

plt.figure(figsize=(10, 6))
plt.plot(x, actual, marker='o', label='Actual Salary')
plt.plot(x, predicted, marker='s', label='Predicted Salary')
plt.title('Comparison of Actual vs. Predicted Salaries')
plt.ylabel('Salary (USD)')
plt.xticks(x)
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```



This is a table comparing the results:

Case	Actual Salary (USD)	Global Model Prediction (USD)	US Model Prediction (USD)
1	80,000	35,120	89,422
2	68,000	48,215	84,173
3	90,300	35,120	105,991
4	60,000	45,375	84,173
5	41,000	45,375	89,422
6	21,000	48,215	89,422
7	125,000	48,215	105,991
8	65,000	35,120	105,991
9	39,520	45,375	84,173
10	86,053	45,375	89,422

Limitations

The limitations of the model mainly come from the data itself, because the survey was an opt-in online survey and this means the participants were mostly white-collar professionals or people active in management and tech communities, while other groups like blue-collar workers or certain industries were underrepresented, so the results may not generalize to every type of workforce. Another issue is the quality of the features, because some of them were very coarse, such as “college degree,” which is just a yes-or-no field that does not consider what someone studied or where they studied, and job titles were written as free text with many unique variations, making them difficult to use directly without better processing like clustering or mapping titles to standard roles. Even though the model tried to address class imbalance, the Medium-income category still had fewer examples and overlapped too much with the Low and High groups, which explains why the model struggled the most with Medium predictions. Finally, survey data always contains noise because people round their salaries, guess, or sometimes intentionally report ranges instead of exact numbers, and some may hesitate to share precise details, which adds even more fuzziness to the dataset, and all of these factors reduce the overall accuracy and reliability of the models..

Ethical Considerations

When it comes to fairness and usage, predicting someone’s salary or income tier is very sensitive, so the model should only be used to uncover possible inequities and never to justify paying someone less, meaning we must avoid statements like “the model says you deserve a lower salary.” Instead, the purpose of the model is to flag cases for human review, such as when an employee is classified in a lower tier than others with similar profiles, which should trigger an investigation into whether that person might be underpaid. There is also a clear bias risk because the model learns from the training data, and if the dataset already contains pay gaps—such as women being paid less on average—the model may unintentionally learn patterns that associate protected characteristics like gender or race with lower pay, and using the model without caution could reinforce these biases. To reduce this risk, sensitive attributes should either be removed or handled with special adjustments, and continuous fairness audits are needed to make sure the model is not quietly disadvantaging any specific group when it is deployed.

Business Implications for GlobaTech

The business implications for GlobaTech show how these models can actually be used in real HR situations, because the classification model can help flag employees who might be underpaid by checking their predicted income tier against their actual salary level, so if a mid-level employee is predicted to fall into the Low tier while they are currently paid in the Medium tier, this difference can alert HR to investigate whether that employee’s pay history or performance suggests they deserve a review or possible raise, and the good thing is that the model almost never misses a truly low-paid

employee due to its very low false-negative rate for the Low tier. The regression model, even though not extremely precise, can still provide rough salary ranges that give clients a general idea of what the expected market pay might be for a certain job profile, such as predicting "X ± Y," and even if Y is large, it still helps indicate whether a company's offer seems too low or too high. The model also showed that industries like tech and higher education levels tend to increase predicted salary, which HR advisors can use when guiding compensation strategies, but GlobaTech should always combine these predictions with expert judgment instead of relying on the model alone. In addition, because the model performed noticeably better in the U.S. than in the global dataset, GlobaTech can use these tools more confidently for U.S.-based audits and planning, while for global or mixed-country analyses the results should be viewed in larger categories such as by region or industry rather than trusting individual predictions, and overall the outputs—including fairness indicators or pay-gap estimates—should always be treated as supportive information rather than the final decision-making tool.

6. Conclusions

It started with a thorough data mining pipeline using the raw data from the salary survey, beginning with cleaning and exploring the dataset to understand the business problem. It was discovered that key factors like years of experience, education level, industry, and geography were major drivers of income differences. Next, cluster analysis was performed, and the data naturally grouped respondents into patterns that reflected career stages and corresponding pay levels, which aligned with the concept of low, medium, and high income tiers. This confirmed that distinct income tiers existed in the data. A decision-tree classifier was then built to categorize employees into Low, Medium, or High income tiers, and it performed well, especially when focused on a single region. At the same time, a random forest regressor was developed to estimate actual salaries, although its predictions showed a high level of uncertainty. Since most responses came from the US and that region offered a more consistent economic context, a separate US only model was created, which turned out to be a strong choice. The US specific model avoided the complications of currency conversion and international economic variation, resulting in more stable and actionable insights for US based workers. For example, this model could clearly identify that a software engineer in San Francisco falls in the High income tier, whereas a global model might misclassify such a profile due to broader variance. These steps provided GlobaTech with data-driven tools that can be applied in practice—for instance, the classification model could be embedded into an HR analytics dashboard to help flag employees whose compensation appears lower than expected for their profile, signaling a possible equity concern or retention risk. The regression model and associated analysis might help with salary benchmarking, enabling firms to identify appropriate pay scales based on experience, industry, and other key characteristics. Such lessons might even enable advisory activities for the clients of GlobaTech, illustrating, for example, that advanced degrees increase salaries by around 20% in some industries, or remote positions in the tech industry tend to pay more. Evidence-based insight supports GlobaTech's value proposition for providing strategic and valuable HR advice. At the same time, limitations of these models were recognized. Future improvements may include collecting more detailed data, such as company tier, regional cost-of-living indexes, and performance metrics. Further exploration of other more advanced machine learning models, such as gradient boosting, or deeper hyper-parameter tuning may be warranted, as time constraints prevented such treatment here. The ability to create additional region-specific models for other major markets, such as Europe or Asia, when enough data has been collected may extend the benefits of the US model to those locations. Ethically, fairness audits are planned to ensure models do not systematically underpredict salaries for protected classes like females or minorities. If such biases are detected, corrective actions, like removing features or adjustments, would be performed. Transparency with clients will continue to be a priority, communicating unequivocally that the intent of these models is to augment, but not replace, decision-making. Concluding, the project represents an example of how machine learning applied to salary survey data can uncover meaningful insights. Although predicting exact salaries remains an ambitious challenge due to the presence of many unmeasured variables, the ability to classify income tiers and highlight influential salary factors holds strong practical value. Importantly, results highlighted the importance of applying models within some consistent context; for example, the US, which aligns with GlobaTech's strategy of tailoring analytics to fit client-specific needs. Using these findings, GlobaTech can continue to enhance its offerings of HR analytics and support firms to develop and maintain fair and competitive pay structures, immediately identify outliers needing attention, and come to a better understanding of compensation trends across roles and regions.

References

Anon, HOW TO USE Matplotlib in 4 MINUTES (2020 Python Tutorial) [online]. *www.youtube.com*. Available at: https://www.youtube.com/watch?v=D4VlmL3G4_o.

Chugh, V., 2023. Python pandas tutorial: The ultimate guide for beginners [online]. *www.datacamp.com*. Available at: https://www.datacamp.com/tutorial/pandas.

Matplotlib, Users guide — Matplotlib 3.5.0 documentation [online]. *matplotlib.org*. Available at: https://matplotlib.org/stable/users/index.html.

Python Programmer, 2019. Learn NUMPY in 5 minutes - BEST Python Library! *YouTube*. Available at: https://www.youtube.com/watch?v=xECXZ3tyONo.

Tech With Tim, 2025. Learn Pandas in 30 Minutes - Python Pandas Tutorial [online]. *YouTube*. Available at: https://www.youtube.com/watch?v=EXIgjIBu4EU [Accessed 10 November 2025].

W3Schools, 2019. Python Tutorial [online]. *W3schools.com*. Available at: https://www.w3schools.com/python/.

Appendix

You may provide further information about your coursework in the appendix. Some examples of the information are

- More tables that describe your results
- More graphics that describe your results